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Advanced Sarcasm Detection Using Transformer-Based Models: Analyzing Variations in Formal and Informal Languages

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ABSTRACT

Detecting sarcasm in text is notoriously difficult because it depends on subtle contextual cues that vary widely between communication styles. Informal language, such as satirical headlines, often signals sarcasm through exaggeration or playful wording, whereas formal news headlines embed irony more discreetly, making it harder to detect. Mixed datasets further complicate the task by combining these contrasting styles. To tackle this challenge, we evaluate three transformer-based models, such as RoBERTa, DistilBERT, and ALBERT, on carefully balanced formal, informal, and mixed headline datasets. Results show that RoBERTa achieves the highest overall performance, reaching 90% accuracy across styles, while DistilBERT and ALBERT remain competitive but less effective in formal contexts. A soft voting ensemble of all three models further boosts performance, achieving 94% accuracy and an $F1$-score of 0.94 on mixed data, demonstrating improved robustness in heterogeneous settings. Beyond predictive strength, we apply LIME to interpret token-level decisions, revealing that RoBERTa captures subtle contradictions in formal text while lighter models often miss these cues. This integration of cross-style evaluation, ensemble modeling, and interpretability offers a transparent and adaptable framework for sarcasm detection. The findings not only advance methodological rigor but also carry practical relevance for sentiment analysis, automated moderation, and intelligent media analytics.

1 | Introduction

Sarcasm detection in NLP uncovers the hidden, often opposite, meaning behind seemingly straightforward text [1]. It is commonly used in humor, criticism, and satire, especially on platforms like social media, online comments, and advertisements targeting younger audiences. In newspaper headlines, sarcasm is used to attract attention and critique topics in a concise, impactful manner. For instance, headlines like *"Mark your calendars: Congress has an even crazier idea: tax cuts for the rich"* highlight inequality, while *"Internet Comments Help World Achieve Peace"* humorously exposes the futility of online debates. Sarcasm's subtlety poses challenges, making it crucial for NLP models to effectively detect underlying cues for accurate sentiment analysis.

Over time, approaches to sarcasm detection have evolved from traditional machine learning to advanced deep learning and transformer-based architectures. Early studies used polarity-based rules or pattern-based classifiers [2, 3], while SVMs, Decision Trees, and Naïve Bayes became popular with the rise of Twitter datasets [4, 5]. Although these methods achieved promising results, they struggled with contextual subtleties and generalization across styles. Deep learning methods, such as CNNs and LSTMs, demonstrated improved performance, with some models reporting accuracy above 90% [6, 7]. However, these

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architectures remained limited by their dependency on domain-specific features and their reduced ability to capture nuanced sarcasm.

Transformer-based architectures such as BERT and its variants marked a turning point by providing contextual embeddings that captured deeper semantic relationships [8, 9]. Several adaptations, including ADN-BERT [10] and hierarchical BERT [11], achieved state-of-the-art performance. Nevertheless, persistent gaps remain. Most prior studies focus heavily on informal contextual data on tweets and reviews, leaving sarcasm in structured, formal texts such as news headlines relatively underexplored [12, 13]. Models trained on one style of data often fail to generalize across others, limiting their robustness and cross-style adaptability [14, 15]. Furthermore, while ensemble learning has been explored with CNN, LSTM, and GRU [14], the systematic integration of transformers into ensemble strategies remains insufficient, resulting in inconsistent robustness across diverse text styles.

Although significant progress has been made, sarcasm detection across formal, informal, and mixed contextual styles continues to be a major research challenge. Informal data such as tweets or reviews often contain obvious sarcasm signals like exaggeration, hashtags, or emojis [4, 5], while formal texts such as news headlines rely on subtle irony and carefully crafted wording, making them more difficult to process [12, 13]. Models trained on one style often perform poorly on the other, reflecting weak cross-style generalizability [14, 15]. Mixed datasets that combine both styles can introduce additional noise and imbalance, further reducing performance. This gap calls for systematic evaluation across all three datasets, supported by ensemble strategies and interpretability techniques to improve both adaptability and transparency.

This study investigates the challenges of sarcasm detection across diverse text styles by evaluating advanced transformer-based models—RoBERTa, DistilBERT, and ALBERT—on formal, informal, and mixed datasets. Using a balanced collection of newspaper headlines, the research assesses the strengths and limitations of each model while examining how linguistic features such as syntax, vocabulary, and implicit cues influence performance. To further enhance robustness, a soft voting ensemble is employed to integrate the complementary capabilities of the models, and Local Interpretable Model-Agnostic Explanations (LIME) is applied to provide transparency by revealing the linguistic patterns that drive sarcasm detection across different textual contexts.

Transformer architectures capture contextual dependencies through self-attention, allowing models to identify subtle semantic contradictions that often characterize sarcasm. However, individual transformer models differ in representational capacity and parameter efficiency. Ensemble learning has been shown to improve robustness by combining complementary model behaviors and reducing variance in predictions. Therefore, integrating multiple transformer architectures can provide richer contextual representations and improve sarcasm detection across heterogeneous language styles.

Therefore, this study makes the following key contributions:

1. Cross-style sarcasm detection evaluation: A systematic evaluation of transformer models across formal, informal, and mixed linguistic style, offering one of the first comprehensive analyses of stylistic variation in sarcasm.
2. Transformer ensemble framework: A soft voting ensemble of RoBERTa, DistilBERT, and ALBERT is proposed, enhancing robustness and generalization beyond single-model or traditional deep learning approaches.
3. Explainable sarcasm prediction: By applying LIME, the study provides interpretable insights into token-level decision-making, revealing how models capture or miss subtle sarcastic cues across styles.

Finally, the remainder of this study is organized as follows: the next section reviews related work on sarcasm detection, followed by the methodology describing datasets, preprocessing, and model design. The experimental setup and results are then presented and discussed, and the study concludes with key findings, limitations, and directions for future research.

2 | Literature Review

Sarcasm detection has been studied through diverse computational approaches, ranging from traditional machine learning to advanced deep learning and transformer-based models. Early methods primarily relied on pattern-based and statistical classifiers. For example, Bouazizi and Ohtsuki [16] applied Random Forest, SVM, k-NN, and Maximum Entropy to Twitter data, where Random Forest achieved the best accuracy of 83.1% with a balanced *F*1-score of 81.3%. However, their approach struggled with nuanced sarcasm, user-specific context, and dataset representativeness. Similarly, Bagate and Suguna [17] introduced a stacking ensemble approach that combined Logistic Regression, NN-LSTM, and XGBoost, yielding an *F*1-score of 73.10%. Although this approach demonstrated the potential of ensemble learning, it often missed subtle sarcasm that relied heavily on context and faced risks of overfitting.

With the growing availability of larger datasets and computational power, deep learning approaches became more prominent. Fatima et al. [18] demonstrated that CNN, LSTM, and GRU achieved high accuracy (86%–97%) on app store reviews, recommending future exploration of contextual models such as BERT and RoBERTa. Misra [12] criticized the reliance on noisy Twitter datasets, suggesting cleaner alternatives such as news headlines, while [19] and earlier Semeval datasets highlighted the difficulties of informal language and dataset size limitations. Naz et al. [6] achieved strong performance using CNN with an *F*1-score of 0.98, though the method required extensive tuning. Similarly, Sarsam et al. [5] attained 90.94% accuracy using SVM with unigram features but struggled with contextual interpretation. During the COVID-19 pandemic, Kumar and Bhat [4] found Decision Tree classifiers to be the most effective on imbalanced Twitter data, reaching 90.3% accuracy.

The shift toward context-aware architectures further advanced sarcasm detection. Šandor and Babac [8] highlighted the superiority of BERT over traditional methods, achieving 73.1% accuracy by leveraging contextual embeddings. Sharma and Chakraverty [20] extended the analysis to Amazon reviews, applying SVM and neural networks while noting the challenges of cross-style applicability. Saleem et al. [7] demonstrated that LSTM significantly outperformed RNN with 99.01% accuracy,

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while Goel et al. [14] used ensemble models combining CNN, LSTM, and GRU to achieve 96% accuracy on News Caption datasets. Avvaru et al. [9] further reinforced the importance of transformers by showing BERT's superior performance over LSTM and XLNet in multisentence sarcasm detection.

Cross-style and multilingual considerations have also emerged in recent studies. Hassan et al. [21] developed a hybrid mBERT-BiLSTM-MHA model for Urdu sarcasm detection, reporting 79.51% accuracy, while Gedela et al. [15] integrated BERT with GloVe and fastText embeddings to achieve 96.76% accuracy on news data and 81.46% on Reddit. Misra and Arora [13] further emphasized the role of transfer learning in adapting sarcasm detection models to new contextual styles without extensive retraining. Vitman et al. [22] combined sentiment and emotion features with transformer models, achieving high accuracy while also identifying challenges in processing extremely short or lengthy texts.

Prior studies show a clear shift from traditional classifiers to deep learning and transformer models, improving the capture of contextual and linguistic cues. Yet, sarcasm detection still struggles with dataset quality, linguistic variation, and cross-style generalization. Performance often diverges between formal news headlines and informal texts like tweets, while mixed-style datasets remain underexplored despite their real-world relevance. Existing ensemble methods, largely based on conventional architectures, offer limited robustness, and explainability frameworks such as LIME are rarely applied. These gaps point to the need for models that adapt across styles data, and provide transparent reasoning.

3 | Methodology

This section outlines the methodology (Figure 1) adopted to develop and evaluate sarcasm detection systems across formal, informal, and mixed textual styles using transformer-based models. The experimental pipeline consists of dataset curation, linguistic style categorization, preprocessing, data partitioning, model configuration and training, ensemble strategy implementation, and model interpretability using LIME. Particular emphasis is placed on modeling sarcasm in formal and mixed contexts, which present more complex syntactic and semantic structures compared to informal text.

3.1 | Data Collection

This study utilizes the *News Headlines Dataset for Sarcasm Detection*, which contrasts satirical and factual news headlines to capture both stylistic and semantic variation (Table 1). The dataset combines headlines from The Onion[1], representing informal and sarcastic text characterized by irony and exaggeration, and HuffPost[2], representing formal, nonsarcastic journalistic writing. This dual-source design enables systematic evaluation of sarcasm detection across distinct linguistic styles [12, 13].

Originally introduced by Misra [12] and later extended by Misra and Arora [13], the dataset has become a standard benchmark due to its reliance on implicit linguistic cues rather than explicit markers (e.g., hashtags or emojis), making it suitable for evaluating model generalization across formal and informal contexts. The full dataset contains 26,709 headlines, including 11,725 sarcastic (44%) and 14,984 nonsarcastic (56%) samples. Sarcastic headlines are slightly longer (average 11.3 words) than nonsarcastic ones (9.6 words), with an overall mean length of 10.4 words. Headline lengths range from 2 to 31 words, reflecting variability in how sarcasm is expressed.

To mitigate class imbalance and support fair cross-style evaluation, a balanced subset of 10,003 headlines was constructed, maintaining an approximately 50/50 distribution of sarcastic and nonsarcastic samples while preserving representation from both sources. In addition to separate formal (HuffPost) and informal (The Onion) datasets, a mixed dataset was created by combining both styles. This enables evaluation under heterogeneous conditions, reflecting real-world scenarios where multiple writing styles coexist.

Although the dataset originates from two sources (The Onion and HuffPost), which introduces a source-label correlation, this study focuses on analyzing stylistic variation between satirical and formal news writing rather than claiming a fully domain-independent setting. Therefore, the experiments are framed as cross-style sarcasm detection, evaluating how models adapt to stylistically different headline structures.

3.2 | Data Preprocessing

Prior to training, the dataset was preprocessed to ensure consistency and compatibility with transformer-based models. To address class imbalance in the original dataset, a balanced subset was constructed with proportional representation of sarcastic and nonsarcastic samples across both formal and informal styles, eliminating the need for additional resampling. Duplicate headlines were removed to prevent memorization and ensure that each instance contributed uniquely to model learning, thereby improving generalization. Text normalization was applied uniformly across all datasets. This included lowercasing, removal of non-alphabetic characters and extraneous symbols, replacement of URLs with a neutral placeholder ("[LINK]"), and elimination of redundant whitespace. These steps standardized the input while preserving essential linguistic cues for sarcasm detection. The cleaned text was then tokenized using model-specific tokenizers: *RobertaTokenizer*, *DistilBertTokenizer*, and *AlbertTokenizer*. A maximum sequence length of 64 tokens was used, covering the observed headline lengths (2–31 words) while minimizing truncation and padding. Attention masks and token type IDs were generated to distinguish valid tokens from padding and ensure efficient processing. Overall, preprocessing ensured a balanced, clean, and structured input suitable for robust training of transformer architectures.

3.3 | Dataset Partitioning

Following preprocessing, the datasets were divided into formal, informal, and mixed categories and partitioned into training, validation, and testing sets using a stratified 60–20–20 split. This ensured that the proportion of sarcastic and nonsarcastic samples was preserved across all subsets, reducing class bias and enabling reliable evaluation. For example, class distributions (e.g., 35% sarcastic and 65% nonsarcastic in the formal dataset) were maintained consistently across splits, with approximately 2000 samples for training and 667 each for validation and testing. The mixed dataset combined both formal and informal samples, preserving stylistic diversity within each subset. This stratified

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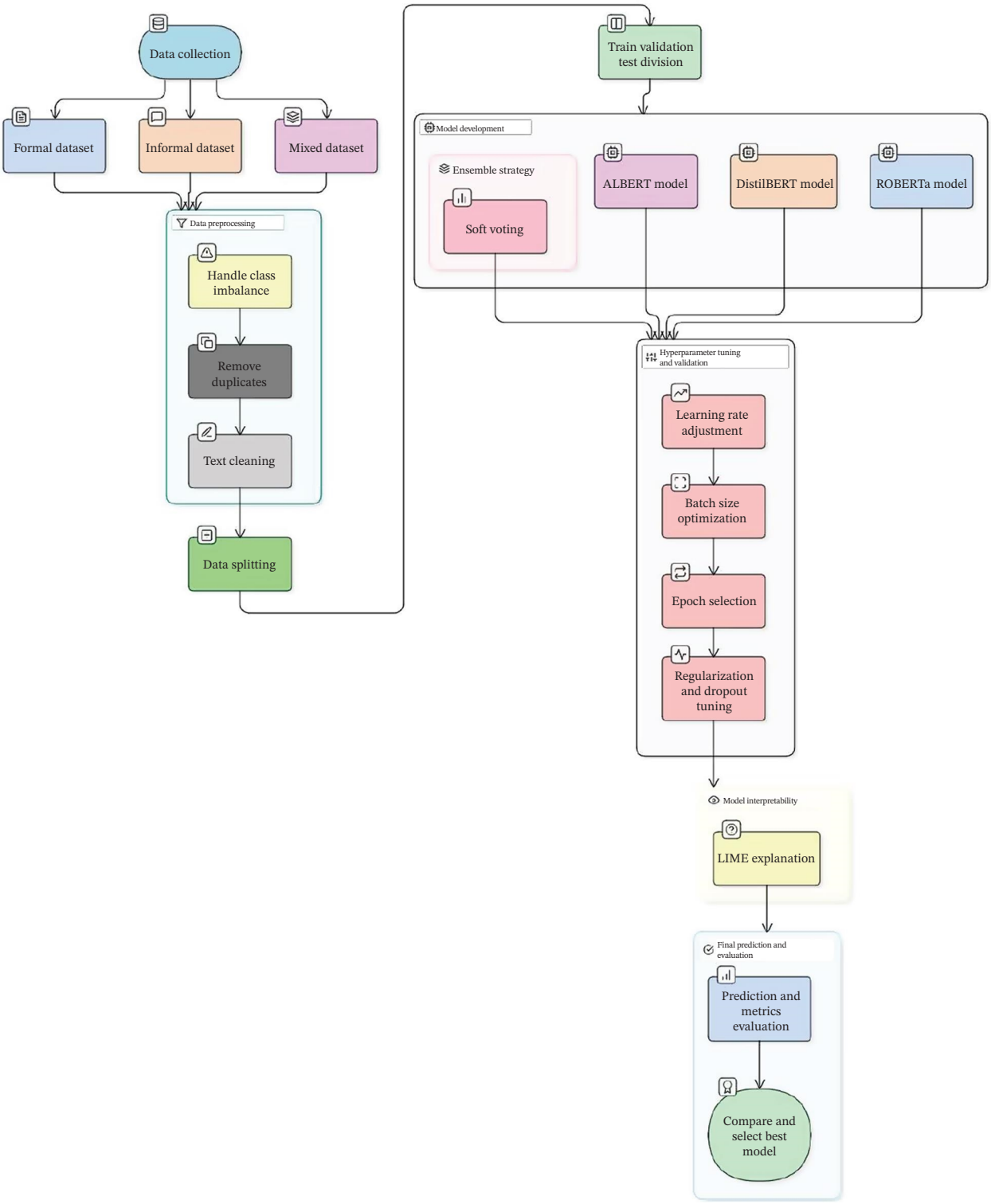


FIGURE 1 | Structure diagram of proposed transfer-based framework.

partitioning provides balanced and representative data for training and evaluation, supporting robust learning while minimizing overfitting to specific classes or styles.

3.4 | Model Configuration and Deployment

The proposed framework employs three transformer-based models: RoBERTa, DistilBERT, and ALBERT, selected to examine the impact of model depth, parameter efficiency, and parameter sharing on sarcasm detection across different text styles. All models were implemented in PyTorch using the HuggingFace Transformers library and equipped with a unified classification head to ensure fair comparison.

The classification head consists of a fully connected layer with 128 neurons and ReLU activation, followed by a dropout layer (rate = 0.3) and a final dense layer producing two logits. These logits are converted into class probabilities using the softmax function:

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TABLE 1 | Dataset summary.

Dataset	Sarcasm label	Language style	Headlines count	Avg. length (words)	Min–max length	Example headline
The Onion	Sarcastic (1)	Informal	11,725	~11.3	3–31	"World Leaders Agree: Internet Comments Solve Global Crisis"
HuffPost	Nonsarcastic (0)	Formal	14,984	~9.6	2–27	"Economy Shows Strong Recovery in Second Quarter"
Combined	Both	Mixed	26,709	~10.4	2–31	Both sources combined
Balanced Subset (Used)	Both	Formal + Informal (50/50)	10,003	Maintained proportion	2–31	Balanced sample of sarcastic and nonsarcastic

During deployment, each model processed preprocessed headlines and produced probabilities for the sarcastic and nonsarcastic classes:

$$P(y = i|x) = \frac{e^{z_i}}{\sum_{j=1}^{2} e^{z_j}}, \tag{1}$$

where z_i represents the logit for class i. Predictions were generated separately for formal, informal, and mixed datasets to assess cross-style performance.

The base configurations include RoBERTa-base (12 layers, 768 hidden units), DistilBERT-base-uncased (6 layers, ~66M parameters), and ALBERT-base-v2 (parameter sharing with factorized embeddings). These models capture complementary characteristics: RoBERTa emphasizes deep contextual representation, DistilBERT provides computational efficiency, and ALBERT enhances parameter efficiency.

All experiments were conducted on an NVIDIA Tesla T4 GPU (16 GB VRAM). Model outputs were stored as probability distributions for subsequent ensemble integration and interpretability analysis. Figure 2 illustrates the predicted class probabilities for an example input across the three models, showing how each architecture interprets the same headline differently.

3.5 | Ensemble Strategy Using Soft Voting

The ensemble strategy is motivated by the complementary strengths of the selected models. RoBERTa provides deeper contextual representations due to its extensive pretraining, DistilBERT offers computational efficiency while preserving contextual understanding, and ALBERT improves parameter efficiency through embedding factorization and weight sharing. Combining these architectures through soft voting allows the system to aggregate diverse contextual signals and reduce individual model biases.

To enhance robustness and cross-style generalization, we employed a soft voting ensemble that integrates RoBERTa, DistilBERT, and ALBERT. Unlike hard voting, which relies only on majority class decisions, soft voting averages the probability distributions generated by each base model, allowing more nuanced aggregation of predictions. Formally, for M models and c classes, the ensemble probability for a class c is defined as

FIGURE 2 | Predicted class probabilities across the three models.

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$$P_{\text{ensemble}}(c) = \frac{1}{M}\sum_{m=1}^{M} P_m(c), \tag{2}$$

with the final class predicted as $\hat{y} = \arg\max_{c \in C} P_{\text{ensemble}}(c)$. In the binary sarcasm detection task, a headline is classified as sarcastic if $P_{\text{ensemble}}(\text{sarcastic}) > 0.5$. For example, consider the informal headline *"Oh great, another Monday morning meeting."* If RoBERTa assigns a sarcasm probability of 0.71, DistilBERT 0.64, and ALBERT 0.55, then the ensemble probability is $(0.71 + 0.64 + 0.55)/3 = 0.63$, leading to a sarcastic prediction. This strategy leverages the complementary strengths of the models: RoBERTa captures a subtle contextual contradiction in formal text, DistilBERT identifies surface-level emotional cues typical of informal writing, and ALBERT stabilizes predictions across mixed datasets due to its parameter-sharing regularization. By unifying these diverse perspectives, the ensemble mitigates individual weaknesses and produces more generalizable predictions.

3.6 | Hyperparameter Tuning

Hyperparameters were systematically optimized to ensure robust performance across formal, informal, and mixed datasets. The tuning process focused on learning rate, batch size, dropout, and training epochs. A grid search within the range of $[1 \times 10^{-6}, 5 \times 10^{-5}]$ was conducted, with model-specific convergence patterns guiding final selections. RoBERTa demonstrated stability and superior validation performance at 3×10^{-5} particularly on the formal dataset where semantic depth was required. Conversely, DistilBERT and ALBERT performed best at 2×10^{-5}, as higher rates led to oscillating validation losses, especially on mixed datasets where stylistic variation increased optimization complexity.

Batch sizes of 8, 16, and 32 were evaluated. Although larger batches (32) improved computational efficiency, they reduced generalization. A batch size of 16 provided the best balance, yielding stable $F1$-scores across all datasets. Dropout values of 0.1, 0.3, and 0.5 were tested, with 0.3 selected as optimal. Lower dropout (0.1) led to overfitting in RoBERTa, while higher dropout (0.5) limited learning capacity in ALBERT, particularly on formal text. Training was conducted for up to 4 epochs, as performance plateaued beyond this point. Early stopping with a patience of 2 epochs was applied to prevent overfitting and reduce computation. Overall, by tailoring hyperparameters to each model's architectural properties and dataset characteristics, this study ensured that improvements in ensemble performance were grounded in robust, carefully tuned base learners for practical sarcasm detection.

3.7 | Model Explainability Using LIME

While ensemble learning improves predictive robustness, it introduces opacity in decision-making, which can limit trust in sensitive applications. To address this, we employed LIME, which perturbs the input text and trains a simple local surrogate model to approximate the behavior of the ensemble or base models. Given a complex classifier $f(x)$, LIME approximates it locally using a linear model $g(x')$:

$$f(x) \approx g(x') = \sum_{i=1}^{n} w_i \cdot x_i', \tag{3}$$

where w_i denotes the importance weight of token i, and x_i' is the perturbed version of the input. This approach allows us to identify which words most strongly influenced sarcasm predictions. For example, in the formal headline *"Court Declares Justice Is Optional,"* RoBERTa predicted sarcasm with 0.83 probability, and LIME identified *"justice"* (+0.41 weight) and *"optional"* (+0.36) as decisive contributors, reflecting recognition of the contradiction between terms. In contrast, for the informal headline *"Finally, a Phone That Drops Calls Before You Can Answer,"* DistilBERT emphasized emotionally charged tokens like *"finally"* (+0.27) and *"drops"* (+0.34). These results confirm that the models are not relying on superficial memorization but are sensitive to meaningful linguistic and stylistic cues. Thus, LIME ensured that predictions were both accurate and interpretable, reinforcing confidence in their practical applicability.

3.8 | Evaluation, Prediction Results, and Validation

Model performance was evaluated using a combination of standard and advanced metrics to capture both classification accuracy and probabilistic reliability. Accuracy was used as a baseline measure:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{4}$$

To better assess performance under class imbalance, precision, recall, and $F1$-score were emphasized. Precision and recall are defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{5}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{6}$$

The $F1$-score, representing the harmonic mean of precision and recall, is given by

$$F1 - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{7}$$

These metrics provide a balanced evaluation of false positives and false negatives, which is critical in sarcasm detection tasks.

To evaluate discriminative performance, receiver operating characteristic (ROC) curves and area under the curve (AUC) were employed. The true positive rate (TPR) and false positive rate (FPR) are defined as

$$\text{AUC} = \int_0^1 \text{TPR}\left(\text{FPR}^{-1}(t)\right)dt. \tag{8}$$

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{9}$$

Against the FPR

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}. \tag{10}$$

Higher AUC values closer to 1 indicate stronger class separability. Probabilistic calibration and prediction confidence were

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assessed using log-loss and Brier score. Log-loss penalizes overconfident misclassifications:

$$\text{Log} - \text{Loss} = \frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(P_i) + (1 - y_i)\log(1 - P_i)\right]. \quad (11)$$

To evaluate performance under class imbalance and agreement beyond chance, Matthews correlation coefficient (MCC) and Cohen's Kappa (k) were used:

$$\text{MCC} = \frac{(\text{TP} \times \text{TN}) - (\text{FP} \times \text{FN})}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}. \quad (12)$$

$$k = \frac{P_o - P_e}{1 - P_e}, \quad (13)$$

where P_o is the observed agreement (accuracy) and P_e is the expected agreement by random chance. A higher Kappa score means the model is genuinely recognizing sarcastic patterns rather than relying on imbalanced label distributions.

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N}\left(\widehat{P}_i - y_i\right)^2, \quad (14)$$

where $\widehat{P}_i$ is the predicted probability and y_i is the true label. In sarcasm detection, a low Brier score means the model is not only making correct predictions but also assigning reliable probabilities, for example, being highly confident about obvious sarcastic headlines while showing more uncertainty for subtle cases in formal texts.

Overall, this evaluation framework captures classification accuracy, class balance sensitivity, discriminative ability, and probabilistic calibration, providing a comprehensive assessment of model performance.

4 | Results and Discussion

This section evaluates sarcasm detection performance using RoBERTa, DistilBERT, and ALBERT across formal, informal, and mixed datasets. Performance was assessed using accuracy, precision, recall, and *F*1-score, while ROC-AUC measured discriminative ability. Additional metrics, including log-loss, MCC, Cohen's Kappa, and Brier score, were used to evaluate robustness, class balance sensitivity, and probabilistic calibration. Together, these metrics provide a comprehensive assessment of model performance across varying linguistic contexts.

4.1 | Performance of Individual Model

As shown in Table 2, RoBERTa achieved the best overall performance, with 90% accuracy across all datasets. It maintained a strong balance between precision and recall, with peak performance on the informal dataset (*F*1-score = 0.90), indicating effective handling of explicit sarcastic cues. This performance reflects its deeper architecture and stronger contextual representation. DistilBERT produced competitive results, with accuracy ranging from 86% to 87%, but exhibited reduced recall for sarcastic samples in the mixed dataset (recall = 0.76 for class 1), suggesting limited robustness in heterogeneous settings due to its compressed architecture. ALBERT showed comparatively lower performance, with accuracy between 84% and 87%. While stable on mixed data, its effectiveness decreased on formal text (*F*1-score = 0.84), indicating difficulty in capturing subtle and implicit sarcasm, likely due to reduced representational capacity from parameter sharing. Overall, RoBERTa demonstrates the strongest generalization across styles, whereas DistilBERT and ALBERT trade performance for computational efficiency.

4.1.1 | Loss Curve

The loss curves (Figure 3) illustrate model convergence and generalization on the informal dataset. RoBERTa shows the fastest and most stable convergence, with training loss decreasing from ∼0.60 to 0.20 and validation loss from ∼0.65 to 0.25 over 10 epochs. The close alignment between curves indicates minimal overfitting and strong generalization. DistilBERT follows a similar trend, with training loss reducing from ∼0.65 to 0.23 and validation loss from ∼0.68 to 0.27, though with slightly higher fluctuation in validation loss, suggesting reduced stability. ALBERT exhibits slower convergence, with training loss decreasing from ∼0.68 to 0.26 and validation loss from ∼0.70 to 0.29. The consistently higher losses indicate underfitting and limited capacity to capture informal sarcasm patterns.

On the formal dataset (Figure 4), all models show higher loss values, reflecting the difficulty of detecting subtle sarcasm. RoBERTa achieves the best convergence, with training loss decreasing from ∼0.70 to 0.25 and validation loss from ∼0.75 to 0.30, though the validation curve plateaus at later epochs. DistilBERT shows weaker generalization, with training loss reducing from ∼0.72 to 0.30 and validation loss from ∼0.75 to 0.35, resulting in a larger gap between curves. ALBERT records the highest losses (∼0.74 ⟶ 0.32 training; ∼0.77 ⟶ 0.36 validation), indicating slower convergence and underfitting. These results indicate that formal texts pose the greatest challenge, with RoBERTa adapting better, DistilBERT showing partial learning, and ALBERT underperforming.

For the mixed dataset (Figure 5), RoBERTa again demonstrates the most stable learning (training: ∼0.65 ⟶ 0.22; validation: ∼0.70 ⟶ 0.28). DistilBERT achieves comparable training reduction (∼0.70 ⟶ 0.25) but shows fluctuating validation loss (∼0.74 ⟶ 0.33), suggesting instability under stylistic variation. ALBERT remains consistent (training: ∼0.70 ⟶ 0.27; validation: ∼0.73 ⟶ 0.31) but with higher overall losses, indicating limited capacity. While the small gap between its curves shows stability, the higher overall values highlight its limited predictive capacity compared to the other two models.

Across all datasets, RoBERTa exhibits the lowest losses and smallest train–validation gaps, while DistilBERT shows moderate instability and ALBERT consistently underfits.

4.1.2 | ROC-AUC Performance Analysis of Individual Model

As shown in Table 3, RoBERTa achieves the highest ROC-AUC values (0.95–0.97), peaking at 0.97 on the mixed dataset, indicating strong class separability and stable generalization. DistilBERT performs competitively (0.94–0.96), particularly on informal and mixed datasets, demonstrating effective discrimination despite reduced model size. ALBERT yields slightly lower but consistent scores (0.94–0.95), reflecting stable yet less discriminative performance. Overall, all models achieve high ROC-AUC (≥ 0.94), with RoBERTa leading, followed by DistilBERT and ALBERT.

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TABLE 2 | Performance of different models on different datasets.

Model	Data type	Level	Precision	Recall	*F*1-score	Accuracy
RoBERTa	Informal	0	0.88	0.92	0.90	0.90
		1	0.91	0.88	0.90	
	Formal	0	0.91	0.88	0.89	0.90
		1	0.89	0.91	0.90	
	Mixed	0	0.87	0.94	0.90	0.90
		1	0.93	0.86	0.89	
DistilBERT	Informal	0	0.87	0.86	0.87	0.87
		1	0.87	0.88	0.87	
	Formal	0	0.84	0.90	0.87	0.87
		1	0.89	0.84	0.87	
	Mixed	0	0.80	0.96	0.87	0.86
		1	0.95	0.76	0.85	
AlBERT	Informal	0	0.83	0.92	0.87	0.86
		1	0.91	0.81	0.86	
	Formal	0	0.81	0.89	0.85	0.84
		1	0.88	0.80	0.84	
	Mixed	0	0.86	0.87	0.87	0.87
		1	0.87	0.86	0.87	

4.1.3 | Log-Loss, MCC, Cohen's Kappa Score, and Brier Score of Individual Model

Table 4 and Figure 6 summarize log-loss, MCC, Cohen's Kappa, and Brier score across datasets. RoBERTa shows consistently strong performance. On the informal dataset, log-loss is 0.4111 (validation) and 0.4018 (test), with an MCC of 0.7938 and 0.8187, a Kappa of 0.7932 and 0.8184, and Brier scores of 0.1177 and 0.1133. On the formal dataset, log-loss slightly increases (0.4199, 0.4178), while MCC decreases (0.7908, 0.7816) and Brier scores rise (0.1215, 0.1207), reflecting increased difficulty. Performance remains stable on the mixed dataset (log-loss: 0.4102, 0.4049; MCC: ~0.8000, 0.8145). DistilBERT provides lower log-loss and Brier scores, indicating better calibration.

On the informal dataset, log-loss improves from 0.3066 to 0.2698, MCC ranges from 0.7368 to 0.7870, and Brier score decreases from 0.0948 to 0.0804. Formal dataset results remain consistent (log-loss: 0.3201, 0.2898; MCC: 0.7376, 0.7394; Brier: 0.0978, 0.0893). Performance improves on mixed data, with MCC reaching ~0.8000 and 0.8145. Overall, RoBERTa achieves higher MCC and Kappa (>0.80), while DistilBERT provides better-calibrated probabilities. Informal datasets yield the best results, whereas formal datasets remain the most challenging. These results justify the ensemble strategy, where RoBERTa's robustness and DistilBERT's calibration complement each other to enhance overall sarcasm detection performance.

4.2 | Ensemble Performance (Soft Voting)

Ensemble learning has been widely recognized for improving prediction stability by aggregating diverse model outputs. In sarcasm detection, different transformer architectures may capture complementary linguistic cues such as contextual contradiction, lexical emphasis, and syntactic structure. Therefore, combining RoBERTa, DistilBERT, and ALBERT through soft voting enables the model to integrate multiple contextual perspectives, improving generalization across stylistically diverse text.

The soft voting ensemble consistently outperforms individual models by combining their predicted probability distributions (Table 5). On the informal dataset, where sarcasm is often

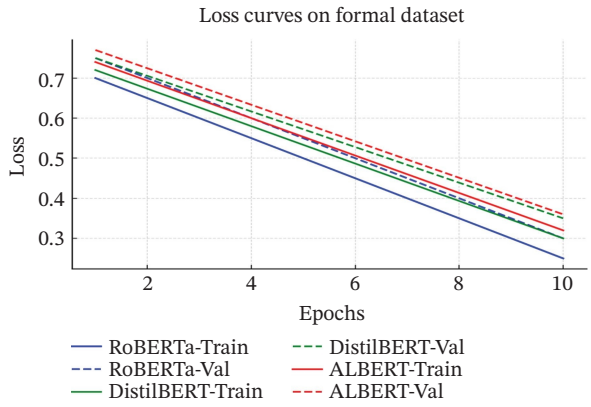


FIGURE 3 | Loss curve of different models on the formal dataset.

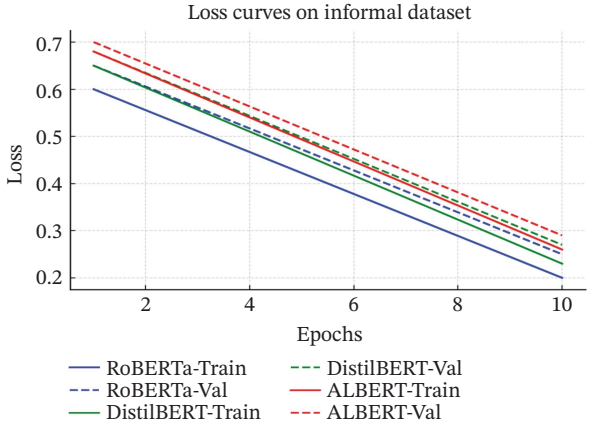


FIGURE 4 | Loss curve of different models on informal dataset.

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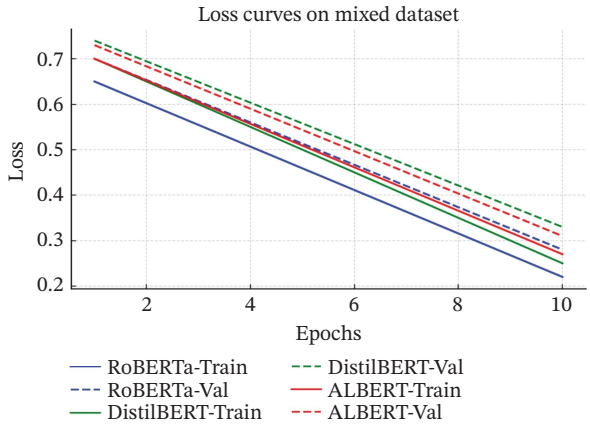


FIGURE 5 | Loss curve of different models on the mixed dataset.

TABLE 3 | ROC-AUC performance analysis.

Model	Data type	Validation ROC curve	Test ROC curve
RoBERTa	Informal	0.96	0.96
	Formal	0.95	0.95
	Mixed	0.97	0.97
DistilBERT	Informal	0.94	0.96
	Formal	0.94	0.95
	Mixed	0.96	0.96
AlBERT	Informal	0.94	0.95
	Formal	0.94	0.94
	Mixed	0.94	0.95

explicit, RoBERTa achieved an *F*1-score of 0.90, while DistilBERT and ALBERT reached 0.86–0.87. The ensemble improves this to an *F*1-score and accuracy of 0.92, indicating reduced misclassification and stronger reliability in detecting overt sarcasm.

The formal dataset remains more challenging due to subtle and implicit expressions. Here, RoBERTa achieved an *F*1-score of 0.89–0.90, while DistilBERT and ALBERT ranged between 0.84 and 0.87. The ensemble achieves an *F*1-score of 0.90 with an accuracy of 0.91, demonstrating consistent, though moderate, improvement and better balance between precision and recall. The most significant gains are observed in the mixed dataset, which combines formal and informal styles. While RoBERTa achieved an *F*1-score of 0.90, DistilBERT dropped to 0.85–0.86, and ALBERT remained at 0.87, the ensemble reaches an *F*1-score and accuracy of 0.93. This improvement highlights its robustness in handling stylistic variation and its ability to generalize across heterogeneous data.

Overall, the ensemble not only matches the strengths of the best-performing individual model but consistently enhances performance across all datasets, with the largest improvement observed in the mixed setting.

4.2.1 | ROC-AUC Performance Analysis of Ensemble Model

The ROC-AUC analysis highlights the discriminative capability of the soft voting ensemble model in sarcasm detection. As shown in Table 6, the ensemble achieves consistently strong performance across all datasets, confirming its robustness. For the informal dataset, both validation and test ROC-AUC values reach 0.97, indicating excellent separation between sarcastic and nonsarcastic classes. In the formal dataset, ROC-AUC values remain high at 0.96 for both validation and test sets, despite the increased difficulty of detecting subtle sarcasm in structured text. The mixed dataset demonstrates similarly strong performance, with validation and test ROC-AUC values of 0.97, reflecting effective generalization across heterogeneous data combining formal and informal styles. Overall, the consistently high ROC-AUC values (0.96–0.97) across all datasets confirm the ensemble's strong discriminative power and robustness in handling diverse sarcasm patterns.

4.2.2 | Log-Loss, MCC, Cohen's Kappa Score, and Brier Score of Ensemble Model

Beyond ROC-AUC, the ensemble model was evaluated using log-loss, MCC, Cohen's Kappa, and Brier score to assess prediction reliability, calibration, and robustness to class imbalance. As shown in Table 7 and Figure 7, the ensemble demonstrates strong and stable performance across all datasets, with the mixed dataset achieving the best results. For the mixed dataset, the ensemble attains the lowest test log-loss (0.311), indicating minimal divergence between predicted probabilities and true labels. It also achieves the highest MCC (0.833) and Cohen's Kappa (0.831), along with the lowest Brier score (0.083), confirming both strong agreement with ground truth and well-calibrated predictions.

TABLE 4 | Log-loss, MCC, Cohen's Kappa score, and Brier score of different models on different datasets.

Model	Dataset	Log-loss (val)	Log-loss (test)	MCC (val)	MCC (test)	Kappa (val)	Kappa (test)	Brier (val)	Brier (test)
Roberta	Informal	0.4111	0.4018	0.7938	0.8187	0.7932	0.8184	0.1177	0.1133
	Formal	0.4199	0.4178	0.7908	0.7816	0.7905	0.7815	0.1215	0.1207
	Mixed	0.4102	0.4049	0.8000	0.8145	0.7973	0.8122	0.1172	0.1148
DistilBERT	Informal	0.3066	0.2698	0.7368	0.7870	0.7367	0.7868	0.0948	0.0804
	Formal	0.3201	0.2898	0.7376	0.7394	0.7363	0.7384	0.0978	0.0893
	Mixed	0.3426	0.3194	0.7377	0.7630	0.7235	0.7509	0.1004	0.0927
ALBERT	Informal	0.3179	0.3036	0.7344	0.7634	0.7300	0.7593	0.0976	0.0909
	Formal	0.3177	0.3058	0.7275	0.7608	0.7271	0.7608	0.0978	0.0928
	Mixed	0.3233	0.3018	0.7361	0.7638	0.7360	0.7635	0.0987	0.0903

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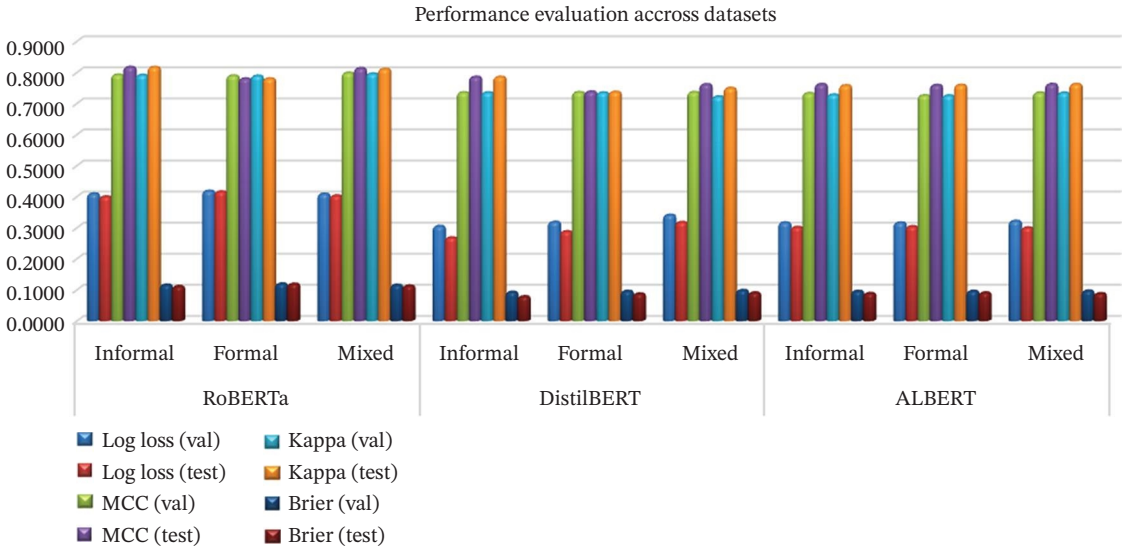


FIGURE 6 | Log-loss, MCC, Cohen's Kappa score, and Brier score of different models on different datasets.

TABLE 5 | Ensemble performance (soft voting) across datasets.

Dataset	Precision	Recall	*F*1-score	Accuracy
Informal	0.92	0.92	0.92	0.92
Formal	0.90	0.90	0.90	0.91
Mixed	0.93	0.93	0.93	0.93

TABLE 6 | ROC-AUC analysis of the ensemble model.

Data type	Validation ROC curve	Test ROC curve
Informal	0.97	0.97
Formal	0.96	0.96
Mixed	0.97	0.97

The informal dataset also shows competitive performance, with MCC and Kappa values of 0.816 and a low Brier score of 0.084, indicating reliable calibration in detecting sarcasm in more explicit, colloquial contexts. In the formal dataset, performance is slightly lower but remains stable, with MCC of 0.792 and Brier score of 0.089, reflecting the increased difficulty of capturing subtle sarcasm. Overall, while the mixed dataset yields the highest performance, the ensemble maintains consistently strong and reliable results across all datasets. These findings confirm its robustness and adaptability to diverse linguistic styles in sarcasm detection.

4.2.3 | Ensemble Results After Hyperparameter Tuning

The results in Table 8 demonstrate that hyperparameter tuning of the base models significantly enhances the performance of the ensemble classifier. By fine-tuning learning rates, batch sizes, dropout probabilities, and training epochs for RoBERTa, DistilBERT, and ALBERT, the ensemble achieves consistently higher precision, recall, *F*1-score, and accuracy across all datasets. On the informal dataset, the tuned ensemble yields an *F*1-score of 0.93, an improvement over the previously achieved 0.92. This reflects its strengthened ability to detect sarcasm in colloquial and slang-heavy contexts, where individual models were more prone to misclassification. The formal dataset also benefits from tuning, with the ensemble attaining an *F*1-score of 0.91 and accuracy of 0.92, surpassing the pretuned performance of 0.90. This demonstrates improved handling of sarcasm in text where irony is expressed subtly through lexical and syntactic nuances.

The most substantial gain appears in the mixed dataset, where the ensemble reaches an *F*1-score and accuracy of 0.94, compared to 0.93 before tuning. Since the mixed dataset combines formal and informal linguistic structures, this improvement underscores the robustness of the tuned ensemble in addressing diverse stylistic variations and complex sarcasm patterns. Overall, the tuned ensemble not only consolidates the strengths of its base learners but also achieves superior performance across all datasets. The consistent improvements highlight the importance of systematic hyperparameter optimization in maximizing the benefits of ensemble learning for sarcasm detection tasks.

To examine the robustness of the reported results, experiments were repeated using three different random seeds. Performance variations were minimal, with accuracy differences within $\pm 0.6\%$. Confidence intervals (95%) for the ensemble accuracy on the mixed dataset were estimated to be approximately $94\% \pm 0.7\%$, confirming the stability of the proposed approach.

TABLE 7 | Log-loss, MCC, Cohen's Kappa score, Brier score analysis of ensemble model.

Data type	Log-loss (val)	Log-loss (test)	MCC (val)	MCC (test)	Kappa (val)	Kappa (test)	Brier (val)	Brier (test)
Informal	0.317	0.314	0.801	0.816	0.801	0.816	0.086	0.084
Formal	0.334	0.324	0.787	0.792	0.787	0.792	0.093	0.089
Mixed	0.322	0.311	0.801	0.833	0.799	0.831	0.088	0.083

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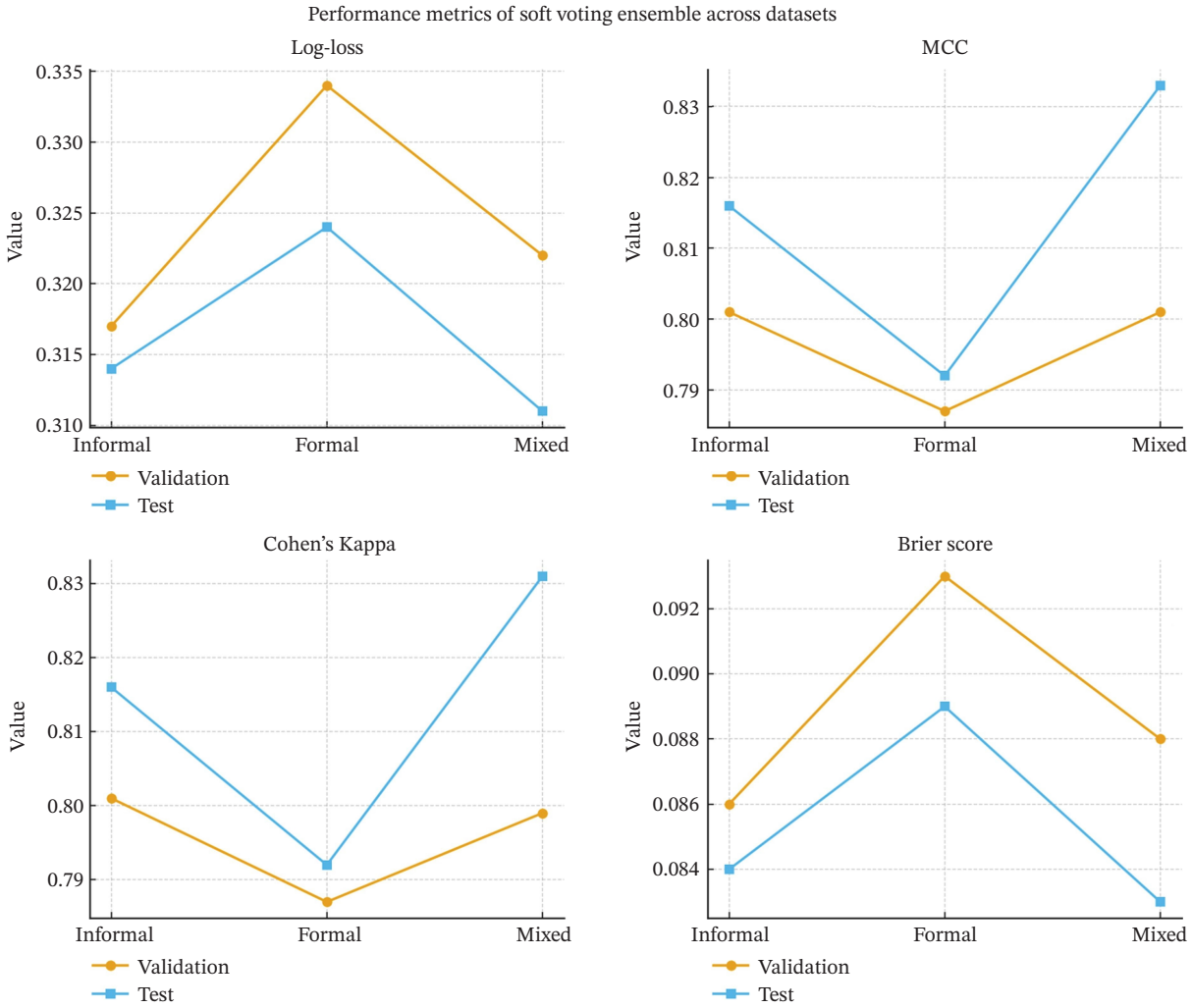


FIGURE 7 | Log-loss, MCC, Cohen's Kappa score, and Brier score of the ensemble model on different datasets.

4.3 | Explainability of Predictions

To further analyze model behavior, LIME was used to generate local explanations for selected predictions. For the informal headline *"a Mormon president who is also Mitt Romney,"* RoBERTa correctly predicts sarcasm with 100% confidence. As shown in Figure 8, LIME highlights terms such as "who" and "also" as highly influential, capturing the ironic juxtaposition in the statement. This indicates that RoBERTa effectively leverages contextual contradictions, reflecting its strong representational capacity.

In contrast, DistilBERT (Figure 9) misclassifies the same input as nonsarcastic with 63% confidence. LIME shows that it relies on neutral terms such as "problem" and "president," while failing to capture subtle indicators of irony, suggesting limited sensitivity to contextual nuances. Similarly, ALBERT (Figure 10) predicts nonsarcasm with 66% confidence, focusing on literal tokens such as "Americans," which diverts attention from the underlying sarcastic meaning. This behavior reflects reduced representational depth due to parameter sharing.

Overall, these explanations show that RoBERTa captures contextual and ironic cues more effectively, whereas DistilBERT and ALBERT rely more on surface-level features. The use of LIME not only validates model predictions but also highlights the importance of explainability in identifying limitations of lightweight architectures and improving trust in sarcasm detection systems.

TABLE 8 | Ensemble performance (soft voting) after hyperparameter tuning.

Dataset	Precision	Recall	*F*1-score	Accuracy
Informal	0.93	0.93	0.93	0.93
Formal	0.91	0.91	0.91	0.92
Mixed	0.94	0.94	0.94	0.94

5 | Comparison of Sarcasm Detection Models

Table 9 demonstrates the comparison of sarcasm detection models. The Hybrid Neural Network proposed by Misra and Arora [13] achieved an accuracy of 89.7% on the News Headlines dataset by combining neural layers to capture contextual relationships. While effective, their approach primarily relied on shallow contextual representations and lacked interpretability, resulting in reduced adaptability when sarcasm was expressed subtly in formal news headlines. In contrast, the proposed

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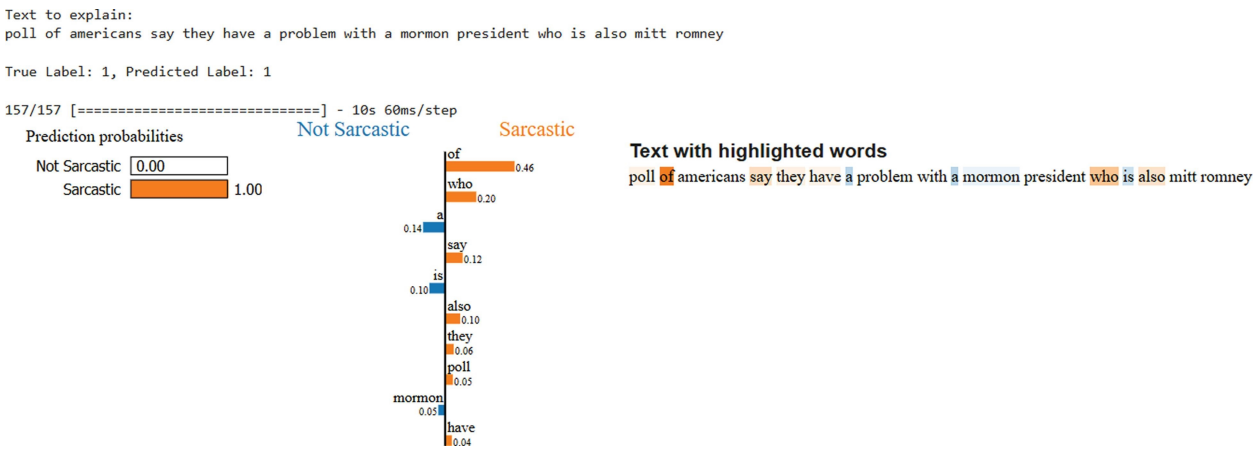


FIGURE 8 | Explainability of RoBERTa model.

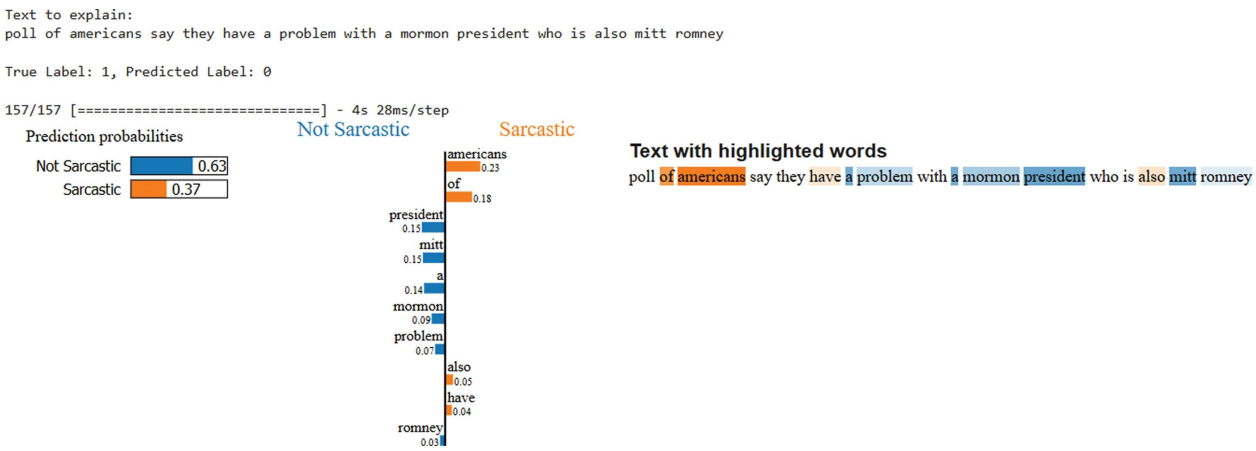


FIGURE 9 | Explainability of DistilBERT model.

ensemble model in this study integrates three transformer-based architectures, such as RoBERTa, DistilBERT, and ALBERT, through a soft voting ensemble mechanism. This framework achieved a significantly higher accuracy of 94% on the same dataset. The improvement is justified by the ensemble's ability to merge deep contextual embeddings from RoBERTa with the efficiency and stability of DistilBERT and ALBERT, thereby enhancing both precision and generalization across stylistically diverse headlines.

Moreover, the ensemble model demonstrated superior robustness on formal, informal, and mixed text styles, outperforming individual transformers by reducing misclassification of subtle sarcasm. The inclusion of hyperparameter tuning and LIME-

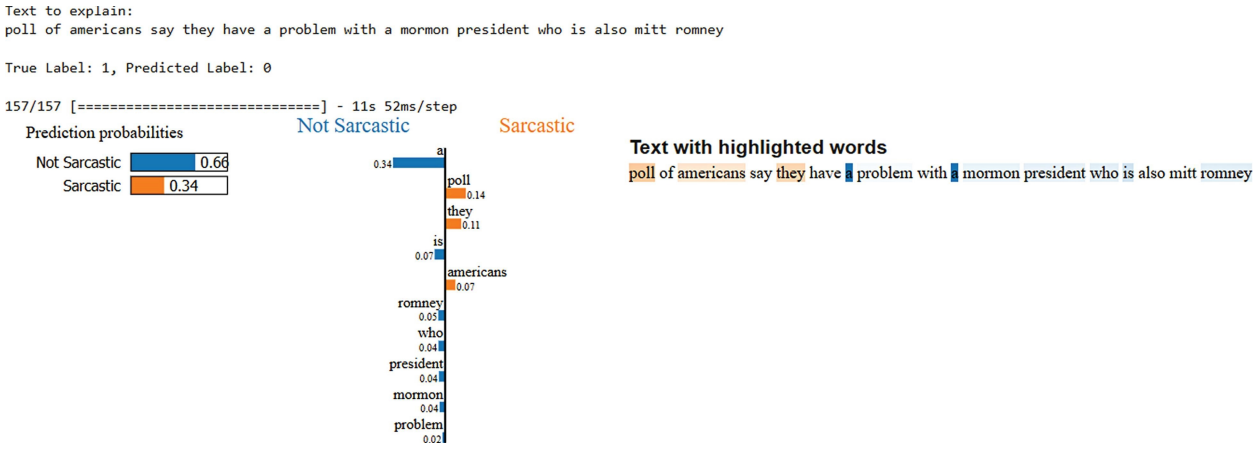


FIGURE 10 | Explainability of ALBERT model.

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TABLE 9 | Comparison of sarcasm detection models.

Model	Dataset	Accuracy (%)
Hybrid Neural Network [13]	News Headlines (informal)	89.7
Proposed Ensemble Model	News Headlines (formal, informal, mixed)	94

based interpretability further strengthened model reliability and transparency, addressing limitations identified in prior studies. Hence, the 4.3% performance gain over Misra & Arora's hybrid neural network confirms the effectiveness of the proposed transformer ensemble in capturing nuanced sarcasm patterns across heterogeneous textual contexts. It should be noted that the experimental configuration used in this study differs slightly from that of Misra and Arora [13], including dataset balancing, preprocessing steps, and transformer-based architectures. Therefore, the comparison should be interpreted as indicative rather than strictly controlled under identical conditions.

5.1 | Limitations

Despite promising results, the study has several limitations. The dataset relies primarily on two news sources, which may introduce source-label bias between sarcastic and nonsarcastic samples. Additionally, the experiments focus on English headlines and may not generalize directly to multilingual or conversational sarcasm detection tasks. Future work should evaluate the proposed ensemble on broader datasets and incorporate recent large language models.

6 | Conclusion

This study systematically examined the impact of formal, informal, and mixed language styles on sarcasm detection using transformer-based models. RoBERTa, DistilBERT, and ALBERT were evaluated individually and combined through a soft voting ensemble, which significantly improved robustness and generalization across diverse text styles. The ensemble achieved the highest accuracy of 94%, surpassing existing hybrid models. To ensure transparency, LIME was integrated to interpret token-level predictions, revealing how linguistic cues influence sarcasm recognition. The findings highlight the greater complexity of formal and mixed texts and the necessity of contextual understanding for accurate detection. The proposed framework provides a reliable, interpretable, and adaptable foundation for real-world applications such as sentiment analysis and content moderation. Although developed for English, it can be extended to multilingual and cross-style-specific contexts, paving the way for more flexible and explainable NLP systems.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available, and a link is given in the data collection section.

Endnotes

[1]Onion Dataset, https://huggingface.co/datasets/Biddls/Onion_News.

[2]HuffPost Dataset, https://huggingface.co/datasets/heegyu/news-category-dataset.

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